Phys 115A HW 3

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Question 1. ***Clever Commutation***
Suppose I give you a quantum system that can be described with a raising operator a^+ and a lowering operator a^- that satisfy the commutation relation $[a^-, a^+] = 1$ (it doesn't have to be the quantum harmonic oscillator, specifically). You know that there is a properly- normalized state ψ that is annihilated by a^-, i.e., you know that $a^-\psi = 0$, but I don't tell you how a^+ acts on ψ.
(a) I prepare for you a state

$$\Psi = c \times (a^-a^-a^-a^-a^+a^+a^+a^+\psi)$$

where c is a real constant. Is Ψ a normalizable state? If so, find c such that Ψ is properly normalized. What is the relation between Ψ and ψ?
(b) I prepare for you a different state

$$\Psi = c \times (a^-a^-a^-a^-a^+a^+\psi)$$

where c is a real constant. Is this new Ψ a normalizable state? If so, find c such that Ψ is properly normalized.

Proof.

(a) Given that $[a^-, a^+] = 1$ (or identity operator), we have $a^-a^+ = 1 + a^+a^-$, so the relation $a^-a^+\psi = \psi + a^+a^-\psi = \psi$, hence ψ is an eigenvector of the linear operator a^-a^+ with eigenvalue 1. Which, if consider the vector $a^+\psi$, we get the following:

$$a^-a^+(a^+\psi) = a^+\psi + a^+(a^-a^+\psi) = a^+\psi + a^+\psi = 2(a^+\psi) \tag{1}$$

Hence, $a^+\psi$ is an eigenvector of a^-a^+ with eigenvalue 2. Inductively, suppose given $n \in \mathbb{N}$ one has $(a^+)^n\psi$ being an eigenvector of a^-a^+ with eigenvalue $(n+1)$, then for the case of $(n+1)$, the following is true:

$$a^-a^+((a^+)^{n+1}\psi) = (a^+)^{n+1}\psi + a^+(a^-(a^+)^{n+1}\psi) = (a^+)^{n+1}\psi + a^+((a^-a^+)(a^+)^n\psi) \tag{2}$$

$$= (a^+)^{n+1}\psi + (n+1)a^+(a^+)^n\psi = (n+2)(a^+)^{n+1}\psi \tag{3}$$

Hence, by induction, all $n \in \mathbb{N}$ satisfies $(a^+)^n\psi$ being an eigenvector of a^-a^+ with eigenvalue $(n+1)$.

With some simplification, another round of induction can be done on the vector of the form $\Psi = (a^-)^n(a^+)^n\psi$: For $n = 2$, one has $(a^-)(a^-a^+)(a^+\psi) = 2a^-a^+\psi = 2\psi$. Which, inductively suppose given $n \in \mathbb{N}$ one has $(a^-)^n(a^+)^n\psi = n!\psi$, then for the case of $(n+1)$, the following is true:

$$(a^-)^{n+1}(a^+)^{n+1}\psi = (a^-)^n(a^-a^+)(a^+)^n\psi = (n+1)(a^-)^n(a^+)^n\psi = (n+1)!\psi \tag{4}$$

Hence, by induction, all $n \in \mathbb{N}$ satisfies $(a^-)^n(a^+)^n\psi = n!\psi$. So, with $\Psi = c(a^-)^4(a^+)^4\psi$, one has $\Psi = c \cdot 4!\psi$, hence with $c! = 0$, Ψ (viewed as a state) is normalizable. With ψ being normalized, it requires $c = \frac{1}{4!}$ for Ψ to be properly normalized.

(b) Adapt the machinery built in part (a), given $\Psi = c(a^-)^4(a^+)^2\psi = c(a^-)^2((a^-)^2(a^+)^2\psi)$, it reduces to $\Psi = c(a^-)^2(2!\psi) = 0$ (since $a^-\psi = 0$). Hence, Ψ is the zero-vector, which is not normalizable.

□

2 (ND, remark (d), calculate (e))

Question 2. ***Constructing States (modified Griffiths 2.10 and 2.14)***
Given the explicit solution for the ground state of the harmonic oscillator,

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

(a) Construct $\psi_2(x)$ explicitly using the algebra formalism. (Hint: $\psi_1(x)$ is constructed this way in Griffiths Example 2.4.)
(b) Check the orthogonality of ψ_0, ψ_1, and ψ_2 by explicit integration. Hint: If you exploit the evenness and oddness of the functions, there is really only one integral left to do.
(c) Find the classical turning points for ψ_0, ψ_1, and ψ_2 (i.e., the values of x where the kinetic energy goes to zero for a classical harmonic oscillator of known total energy).
(d) Sketch ψ_0, ψ_1, ψ_2, ψ_3. For each state, mark the classical turning point. How does the classical turning point relate to the form of the wavefunction?
(e) For ψ_0, what is the probability (correct to three significant digits) of finding the particle outside the classically allowed region, meaning outside the classical turning points? You can use a math table under "Normal Distribution" or Mathematica or any other tool you wish to numerically evaluate the relevant integral.

Proof. For this question, recall the definition of the operators: $a_\pm = \frac{1}{\sqrt{2\hbar mw}}(mw\hat{x} \mp i\hat{p})$, $\hat{x}$ is the operator simply by multiplying x, and $\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}$. So, $a_\pm = \frac{1}{\sqrt{2\hbar mw}}(mwx \mp \hbar\frac{\partial}{\partial x})$.

(a) Recall that $\psi_n(x) = \frac{1}{\sqrt{n!}}(a_+)^n\psi_0(x)$. So, for $n = 2$, one derives the following:

$$a_+\psi_0(x) = \frac{1}{\sqrt{2\hbar mw}}\left(mwx - \hbar\frac{\partial}{\partial x}\right)\left(\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}\right) \tag{5}$$

$$= \sqrt{\frac{mw}{2\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} + \sqrt{\frac{\hbar}{2mw}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} \frac{mw}{\hbar}xe^{-\frac{mw}{2\hbar}x^2} \tag{6}$$

$$= 2\sqrt{\frac{mw}{2\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} = \sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} \tag{7}$$

$$(a_+)^2\psi_0(x) = \frac{1}{\sqrt{2\hbar mw}}\left(mwx - \hbar\frac{\partial}{\partial x}\right)\left(\sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2}\right) \tag{8}$$

$$= \frac{mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2 e^{-\frac{mw}{2\hbar}x^2} - \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2} + \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} \cdot \frac{mw}{\hbar}x^2 e^{-\frac{mw}{2\hbar}x^2} \tag{9}$$

$$= \frac{2mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2 e^{-\frac{mw}{2\hbar}x^2} - \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2} \tag{10}$$

So, we get that $\psi_2(x) = \frac{1}{\sqrt{2}}(a_+)^2\psi_0(x) = \frac{\sqrt{2}mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2 e^{-\frac{mw}{2\hbar}x^2} - \frac{1}{\sqrt{2}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}$.

(b) It suffices to check orthogonality for $\psi_0, (a_+)\psi_0, (a_+)^2\psi_0$ (since scaling doesn't affect orthogonality).

For $\psi_0, (a_+)\psi_0$, the integration provides the following:

$$\int_{-\infty}^{\infty} ((a_+)\psi_0(x))^*\psi_0(x)dx = \int_{-\infty}^{\infty} \sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} \cdot \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2} dx = 0 \tag{11}$$

(Note: The above is an integration of an odd function that converges, hence provides 0). Hence, $\psi_0, (a_+)\psi_0$ are orthogonal under integration of inner product.

For $(a_+)\psi_0, (a_+)^2\psi_0$, the integration provides the following:

$$\int_{-\infty}^{\infty} ((a_+)\psi_0(x))^*((a_+)^2\psi_0(x))dx \tag{12}$$

$$= \int_{-\infty}^{\infty} \sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} \cdot \left(\frac{\sqrt{2}mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2 e^{-\frac{mw}{2\hbar}x^2} - \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}\right) dx \tag{13}$$

$$= K\int_{-\infty}^{\infty} x^3 e^{-\frac{mw}{\hbar}x^2} dx + L\int_{-\infty}^{\infty} xe^{-\frac{mw}{\hbar}x^2} dx = 0 \tag{14}$$

(Note: Above are two odd function that has converging integrals, which with domain being symmetric it converges to 0. K, L are proper substitutions of the constants). Hence, $(a_+)\psi_0, (a_+)^2\psi_0$ are also orthogonal under integration inner product.

Finally, for $\psi_0, (a_+)^2\psi_0$, the integration provides the following:

$$\int_{-\infty}^{\infty} (\psi_0(x))^*((a_+)^2\psi_0(x))dx \tag{15}$$

$$= \int_{-\infty}^{\infty} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}\left(\frac{\sqrt{2}mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2 e^{-\frac{mw}{2\hbar}x^2} - \frac{1}{\sqrt{2}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}\right) dx \tag{16}$$

$$= -\frac{1}{\sqrt{2}}\int_{-\infty}^{\infty} \sqrt{\frac{mw}{\pi\hbar}} e^{-(\sqrt{\frac{mw}{\hbar}}x)^2} dx + \frac{\sqrt{2}mw}{\hbar}\int_{-\infty}^{\infty} \sqrt{\frac{mw}{\pi\hbar}} x^2 e^{-\frac{mw}{\hbar}x^2} dx \tag{17}$$

$$= -\frac{1}{\sqrt{2}} - \frac{\sqrt{2}mw}{\hbar}\left(\int_{-\infty}^{\infty} \sqrt{\frac{mw}{\pi\hbar}}\frac{\partial}{\partial\lambda}\left(e^{-\lambda x^2}\right) dx\right)\Bigg|_{\lambda=\frac{mw}{\hbar}} \tag{18}$$

$$= -\frac{\sqrt{2}mw}{\hbar}\frac{d}{d\lambda}\Bigg|_{\lambda=\frac{mw}{\hbar}} \int_{-\infty}^{\infty} \sqrt{\frac{mw}{\pi\hbar}} e^{-\lambda x^2} dx - \frac{1}{\sqrt{2}} = -\frac{\sqrt{2}mw}{\hbar}\sqrt{\frac{mw}{\pi\hbar}}\sqrt{\frac{\pi}{\lambda}}\Bigg|_{\lambda=\frac{mw}{\hbar}} - \frac{1}{\sqrt{2}} \tag{19}$$

$$= -\sqrt{\frac{2}{\pi}}\left(\frac{mw}{\hbar}\right)^{\frac{3}{2}} \cdot \left(-\frac{1}{2}\frac{\sqrt{\pi}}{\lambda^{\frac{3}{2}}}\right)\Bigg|_{\lambda=\frac{mw}{\hbar}} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0 \tag{20}$$

Hence, under integration inner product, $\psi_0, (a_+)^2\psi_0$ are also orthogonal.

Since ψ_0, ψ_1, ψ_2 are scalar multiples of $\psi_0, (a_+)\psi_0, (a_+)^2\psi_0$ respectively, hence they're also mutually othogonal under integration inner product.

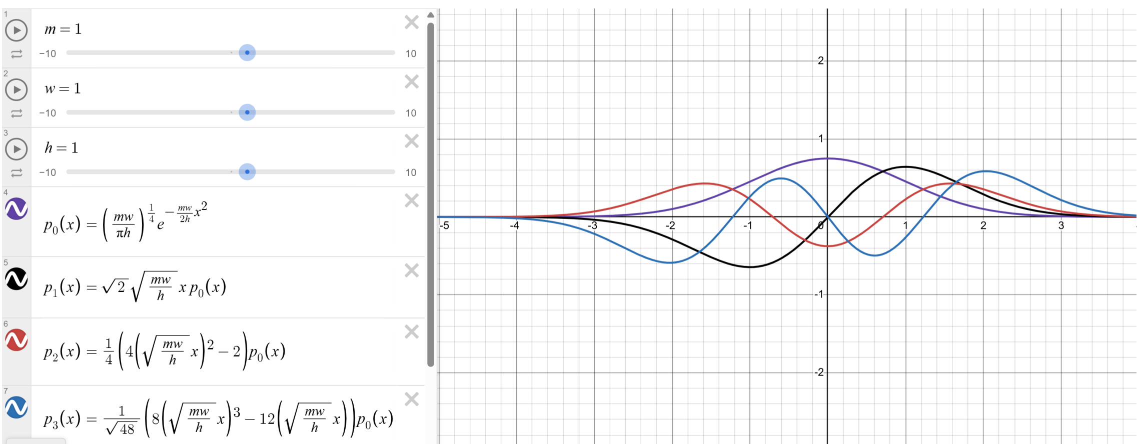
(c) Given the energy of $\psi_n(x)$ is $E_n = \left(n+\frac{1}{2}\right)\hbar w$, and the potential energy is $V(x) = \frac{1}{2}mw^2x^2$, the classical turning points occurs when $E_n = V(x)$, providing the following solution:

$$\frac{1}{2}mw^2x^2 = \left(n+\frac{1}{2}\right)\hbar w \implies x = \pm\sqrt{\left(n+\frac{1}{2}\right)\frac{2\hbar}{mw}} = \pm\sqrt{\frac{(2n+1)\hbar}{mw}} \tag{21}$$

Hence, the classical turning points of the four states ψ_0, ψ_1, ψ_2 (denoted as $\pm x_0, \pm x_1, \pm x_2$ respectively) are given as:

$$\pm x_0 = \pm\sqrt{\frac{\hbar}{mw}}, \quad \pm x_1 = \pm\sqrt{\frac{3\hbar}{mw}}, \quad \pm x_2 = \pm\sqrt{\frac{5\hbar}{mw}} \tag{22}$$

(d) Here are some graphs of the first four states (starting from ground states):



Here, the classical turning point actually represents the farthest positions from the origin where the second derivative of the wavefunction is 0 (or the farthest point the wave function changes its concavity).

(e) Recall that from part (d), the classical turning point

$\square$

3 (ND, tweak constants in (a))

Question 3. ***Uncertainty Principle for the nth state (modified Griffiths 2.12)***
Consider the nth stationary state of the harmonic oscillator.
(a) Find $\langle x\rangle$, $\langle p\rangle$, $\langle x^2\rangle$ and $\langle p^2\rangle$ for this state, following the method of Example 2.5.
(b) Find the expectation value of the kinetic energy, $\langle T\rangle$, and the expectation value of the potential energy, $\langle V\rangle$, for this state.
(c) Compute $\sigma_x\sigma_p$ for this state, and check that the uncertainty principle is satisfied. Under what circumstances is it saturated (i.e., is $\sigma_x\sigma_p = \hbar/2$)?
Hint: For this problem, do not explicitly integrate the wavefunction, which has a terrible form for arbitrary n. Instead, notice that you can write x and p in terms of raising and lowering operators,

$$x = \sqrt{\frac{\hbar}{2m\omega}}(a^+ + a^-), \quad p = i\sqrt{\frac{\hbar m\omega}{2}}(a^+ - a^-).$$

Then to compute the expectation values you just have to act on ψ_n with raising and lowering operators, which you know how to do.

Proof. Here, we'll assume the fact that the list $\{\psi_n\}_{n\in\mathbb{N}}$ forms an orthonormal list under integration inner product (so, for $n \neq m$, it satisfies $\int_{-\infty}^{\infty}\psi_n(x)^*\psi_m(x)dx = 0$). Also, we'll utilize the fact that a^+, a^- are Adjoint Operators/Hermition Conjugates of each other, which implies that $\int_{-\infty}^{\infty}\psi(x)^*(a^+\phi(x))dx = \int_{-\infty}^{\infty}(a^-\psi(x))^*\phi(x)dx$, and $\int_{-\infty}^{\infty}\psi(x)^*(a^-\phi(x))dx = \int_{-\infty}^{\infty}(a^+\psi(x))^*\phi(x)dx$.

Also, based on these rule, let T^* denote the adjoint / Hermitian Conjugate of linear operator T, then we get that $\hat{p}^* = -i\sqrt{\frac{\hbar mw}{2}}((a^+)^* - (a^-)^*) = -i\sqrt{\frac{\hbar mw}{2}}(a^- - a^+) = i\sqrt{\frac{\hbar m\omega}{2}}(a^+ - a^-) = \hat{p}$.

(a) Given $\psi_n(x) = (a^+)^n\psi_0(x)$, using the expression that $\hat{x} = \sqrt{\frac{\hbar}{2mw}}(a^+ + a^-)$ as operator, we get the following (where below is up to some constant):

$$\langle x\rangle = \int_{-\infty}^{\infty}\psi_n(x)^*(\hat{x}\psi_n(x))dx = \sqrt{\frac{\hbar}{2mw}}\int_{-\infty}^{\infty}((a^+)^n\psi_0(x))^*((a_+)^{n+1}\psi_0(x) + a^-(a^+)^n\psi_0(x))dx \tag{23}$$

$$= \sqrt{\frac{\hbar}{2mw}}\int_{-\infty}^{\infty}\psi_n(x)^*\psi_{n+1}(x)dx + \sqrt{\frac{\hbar}{2mw}}\int_{-\infty}^{\infty}((a^+)^n\psi_0(x))^*a^-((a^+)^n\psi_0(x))dx \tag{24}$$

$$= 0 + \sqrt{\frac{\hbar}{2mw}}\int_{-\infty}^{\infty}((a^+)^{n+1}\psi_0(x))^*\psi_n(x)dx = \sqrt{\frac{\hbar}{2mw}}\int_{-\infty}^{\infty}\psi_{n+1}(x)^*\psi_n(x)dx = 0 \tag{25}$$

$$\langle p\rangle = \int_{-\infty}^{\infty}\psi_n(x)^*(\hat{p}\psi_n(x))dx = i\sqrt{\frac{\hbar mw}{2}}\int_{-\infty}^{\infty}((a^+)^n\psi_0(x))^*((a^+)^{n+1}\psi_0(x) - a^-(a^+)^n\psi_0(x))dx \tag{26}$$

$$= i\sqrt{\frac{\hbar mw}{2}}\left(\int_{-\infty}^{\infty}\psi_n(x)^*\psi_{n+1}(x)dx - \int_{-\infty}^{\infty}((a^+)^{n+1}\psi_0(x))^*((a^+)^n\psi_0(x))dx\right) \tag{27}$$

$$= 0 - i\sqrt{\frac{\hbar mw}{2}}\int_{-\infty}^{\infty}\psi_{n+1}(x)^*\psi_n(x)dx = 0 \tag{28}$$

$$
\begin{aligned}
\langle x^2 \rangle &= \int_{-\infty}^{\infty} \psi_n(x)^2 x^2 \psi_n(x) dx = \int_{-\infty}^{\infty} (\hat{x}\psi_n(x))^* (\hat{x}\psi_n(x)) dx && (29)\\
&= \frac{\hbar}{2mw} \int_{-\infty}^{\infty} ((a^+)^{n+1}\psi_0(x) + a^-(a^+)^n\psi_0(x))^*((a^+)^{n+1}\psi_0(x) + a^-(a^+)^n\psi_0(x)) dx && (30)\\
&= \frac{\hbar}{2mw} \int_{-\infty}^{\infty} (\sqrt{(n+1)!}\psi_{n+1}(x) + \sqrt{n!}\psi_{n-1}(x))^*(\sqrt{(n+1)!}\psi_{n+1}(x) + \sqrt{n!}\psi_{n-1}(x)) dx && (31)\\
&= \frac{\hbar}{2mw}((n+1)! + n!) && (32)
\end{aligned}
$$

(Note: the last equality is using the fact that the integration represents the $\|\sqrt{(n+1)!}\psi_{n+1}(x) + \sqrt{n!}\psi_{n-1}(x)\|^2$, by orthonormality and Pythagoreus Theorem of inner product, such norm is the same as $\|\sqrt{(n+1)!}\|\psi_{n+1}(x)\|^2 + \|\sqrt{n!}\psi_{n-1}(x)\|^2 = (n+1)! + n!$).

$$
\begin{aligned}
\langle p^2 \rangle &= \int_{-\infty}^{\infty} \psi_n(x)^* (\hat{p}^2\psi_n(x)) dx = \int_{-\infty}^{\infty} (\hat{p}\psi_n(x))^* (\hat{p}\psi_n(x)) dx && (33)\\
&= \frac{\hbar mw}{2} \int_{-\infty}^{\infty} (a^+\psi_n(x) - a^-\psi_n(x))^*(a^+\psi_n(x) - a^-\psi_n(x)) dx && (34)\\
&= \frac{\hbar mw}{2}(2n+1) && (35)
\end{aligned}
$$

(check the precise statement about how a^+, a^- acts on $\psi_n(x)$ with constant).

(b) Kinetic energy $\hat{T} := \frac{\hat{p}^2}{2m}$, hence $\langle T \rangle = \int_{-\infty}^{\infty} \psi_n(x)^* (\frac{\hat{p}^2}{2m}\psi_n(x)) dx = \frac{\langle p^2 \rangle}{2m}$.

Similarly, given potential energy $V(x) = \frac{1}{2}mw^2x^2$, then $\langle V \rangle = \int_{-\infty}^{\infty} \psi_n(x)^* (\frac{1}{2}mw^2x^2\psi_n(x)) dx = \frac{1}{2}mw^2 \langle x^2 \rangle$.

(c) Need to wait until the constants in (a) is checked.

$\square$

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Question 4. ***Back of the envelope***
Using the symmetry of the potential, come up with an expression for the zero-point energy of a harmonic oscillator on the assumption that it is the minimum energy required by the uncertainty principle.

Proof. Recall that Uncertainty Principle states that $\sigma_p\sigma_x \geq \frac{\hbar}{2}$, where $\sigma_j := \sqrt{\langle j^2 \rangle - \langle j \rangle^2}$ for operator $j = p, x$. Hence, it must satisfy the following equation by squaring these quantities:

$$\left(\langle p^2 \rangle - \langle p \rangle^2\right)\left(\langle x^2 \rangle - \langle x \rangle^2\right) \geq \frac{\hbar^2}{4} \tag{36}$$

Now, assume that ψ is a stationary state for Simple Harmonic Oscillator, which with potential $V(x) = \frac{1}{2}mw^2x^2$, it satisfies the stationary Schrödinger equation for energy $E \geq 0$ as follow:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}mw^2x^2\psi = E\psi \implies \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = \left(\frac{1}{2}mw^2x^2 - E\right)\psi \tag{37}$$

Hence, we get the following for $\langle p^2 \rangle$ (recall that $\hat{p} := \frac{\hbar}{i}\frac{\partial}{\partial x}$ and $\hat{x} := x$ as operator):

$$\begin{aligned}
\langle p^2 \rangle &= \int_{-\infty}^{\infty} \psi^* \left(\hat{p}^2\psi\right) dx = -\int_{-\infty}^{\infty} \psi^* \left(\hbar^2\frac{\partial^2\psi}{\partial x^2}\right) dx = -2m\int_{-\infty}^{\infty} \psi^* \left(\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}\right) dx &(38)\\
&= -2m\int_{-\infty}^{\infty} \psi^* \left(\frac{1}{2}mw^2x^2 - E\right)\psi dx = 2mE\int_{-\infty}^{\infty} \psi^*\psi dx - m^2w^2\int_{-\infty}^{\infty} \psi^* \cdot (\hat{x}^2\psi) dx &(39)\\
&= 2mE - m^2w^2\langle x^2 \rangle &(40)
\end{aligned}$$

Also, recall that the states ψ can be chosen as even or odd, while each are chosen to have real values, based on what was proven in **HW 2**. Let $\psi_{\text{odd}}, \psi_{\text{even}} : \mathbb{R} \to \mathbb{R}$ denote the even / odd case of the wave function respectively. Then, since derivative of odd/even function becomes even/odd function respectively, while x is an odd function. Hence, $\psi^*_{\text{odd}}(\hat{p}\ \psi_{\text{odd}}) = \frac{\hbar}{i}\psi_{\text{odd}}\frac{\partial\psi_{\text{odd}}}{\partial x}$, $\psi^*_{\text{odd}}(x\psi_{\text{odd}})$, $\psi^*_{\text{even}}(\hat{p}\ \psi_{\text{even}}) = \frac{\hbar}{i}\psi_{\text{even}}\frac{\partial\psi_{\text{even}}}{\partial x}$, and $\psi^*_{\text{even}}(x\psi_{\text{even}})$ are all odd functions. Hence, with ψ chosen to be odd / even, we have $\langle p \rangle, \langle x \rangle = 0$ (since $\int_{-\infty}^{\infty} \psi^*(\hat{p}\psi)dx = 0 = \int_{-\infty}^{\infty} \psi^*(x\psi)dx$ due to the function always being odd, based on the choice).

Now, let $y := \langle x^2 \rangle$, with the above machinery and the first equation based on Uncertainty Principle, we derived the following:

$$\begin{aligned}
\left(\langle p^2 \rangle - \langle p \rangle^2\right)\left(\langle x^2 \rangle - \langle x \rangle^2\right) = (2mE - m^2w^2y)y \geq \frac{\hbar^2}{4} &\quad(41)\\
\implies m^2w^2 \cdot y^2 - 2mE \cdot y + \frac{\hbar^2}{4} \leq 0 &\quad(42)
\end{aligned}$$

Since the leading coefficient $m^2w^2 > 0$ can be assumed, the above inequality is equivalent to having the above quadratic polynomial having at least 1 real solution (intuitively/graphically, since it's concaving up, with some points ≤ 0, it must passes through the horizontal axis at least once).

Hence, the discriminant $b^2 - 4ac \geq 0$, showing that $4m^2E^2 - m^2w^2\hbar^2 \geq 0$, or $E^2 \geq \frac{w^2\hbar^2}{4}$; with $E \geq 0$, it must satisfy $E \geq \frac{w\hbar}{2}$. So, $E \geq \frac{w\hbar}{2}$ is the minimum energy required to satisfy Uncertainty Principle. ☐

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Question 5. ***Coherent states***

In problem 3, you found that most of the stationary states of the harmonic oscillator satisfy, but do not saturate, the uncertainty principle. Now let us consider a special superposition of these stationary states called a coherent state, which has some fairly interesting properties. We call such a state an eigenfunction of the lowering operator. It just means the coherent state $S_\alpha(x)$ *satisfies* $a^- S_\alpha(x) = \alpha S_\alpha(x)$ *where* α *is a complex number. Note that there are no normalizable eigenfunctions of the raising operator.*

(a) Calculate $\langle x \rangle$ *and* $\langle p \rangle$ *for the state* $S_\alpha(x)$*. Does this remind you of anything from previous problems in this set? Hint: Since you know how* a^- *acts on* $S_\alpha(x)$*, use the fact that*

$$\int_{-\infty}^{\infty} f^*(x)(a^{\pm} g(x))\, dx = \int_{-\infty}^{\infty} (a^{\mp} f(x))^* g(x)\, dx.$$

Also, keep in mind that α *is in general complex.*

(b) Calculate $\langle x^2 \rangle$ *and* $\langle p^2 \rangle$ *for the state* $S_\alpha(x)$*.*

(c) Show that S_α *saturates the uncertainty principle.*

(d) We can express $S_\alpha(x)$ *at* $t = 0$ *explicitly in terms of the stationary states* ψ_n*,*

$$S_\alpha(x) = \sum_{n=0}^{\infty} c_n \psi_n(x)$$

Show that $c_n = \frac{\alpha^n}{\sqrt{n!}} c_0$*. Hint: Keep using the earlier hint!*

(e) Fix c_0 *by normalizing* $S_\alpha(x)$*.*

(f) Now consider $S_\alpha(x,t)$*,*

$$S_\alpha(x,t) = \sum_{n=0}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

Show that $S_\alpha(x,t)$ *remains an eigenfunction of* a^- *but that* α *now evolves in time. That is, show that* $a^- S_\alpha(x,t) = \alpha(t) S_\alpha(x,t)$*, where* $\alpha(t) = e^{-i\omega t}\alpha$*. Argue that this means coherent states stay coherent in time, and continue to saturate the uncertainty principle.*

(g) Can we think of the ground state ψ_0 *as a coherent state? What is* α *in this case?*

These states are profoundly interesting in quantum optics, since they provide an accurate quantum description of laser light. Roy Glauber was awarded half of the Nobel Prize in Physics in 2005 for his 1963 work on understanding the properties of coherent states in quantum optics.

Proof. First, recall the following definition of $\hat{x}, \hat{p}$ using ladder operators a^+, a^-, together with their commutation relation:

$$\hat{x} = \sqrt{\frac{\hbar}{2mw}}(a^+ + a^-), \quad \hat{p} = i\sqrt{\frac{\hbar m w}{2}}(a^+ - a^-), \quad [a^-, a^+] = a^- a^+ - a^+ a^- = \mathrm{id} \tag{43}$$

Which, $\hat{x}^2, \hat{p^2}$ respectively can also be expanded as follow:

$$\hat{x}^2 = \frac{\hbar}{2mw}((a^+)^2 + a^+a^- + a^-a^+ + (a^-)^2) = \frac{\hbar}{2mw}((a^+)^2 + \mathrm{id} + 2a^+a^- + (a^-)^2) \tag{44}$$

$$\hat{p}^2 = -\frac{\hbar m w}{2}((a^+)^2 - a^+a^- - a^-a^+ + (a^-)^2) = -\frac{\hbar m w}{2}((a^+)^2 - \mathrm{id} - 2a^+a^- + (a^-)^2) \tag{45}$$

Also, let $\alpha = u + iv$ as complex number (where $u = \text{Re}(\alpha)$ and $v = \text{Im}(\alpha)$), we have $|\alpha|^2 = u^2 + v^2$, and $\alpha^2 = (u^2 - v^2) + i(2uv)$, so $\text{Re}(\alpha^2) = u^2 - v^2$ and $\text{Im}(\alpha^2) = 2uv$.

Rmk: Before starting, we'll use standard inner product notation $\langle f, g\rangle := \int_{-\infty}^{\infty}(g(x))^* f(x)dx$ for easier notation. Hence, the operator $\hat{x}, \hat{p}$ has the expectation values $\langle x\rangle = \langle \hat{x}\psi, \psi\rangle$ and $\langle \hat{p}\psi, \psi\rangle$; also, with a^-, a^+ being adjoints / Hermitian Conjugates of each other, the inner product notation becomes: $\langle a^- f, g\rangle = \langle f, a^+ g\rangle$ and $\langle a^+ f, g\rangle = \langle f, a^- g\rangle$.

(a) Given $a^- \ S_\alpha(x) = \alpha \cdot S_\alpha(x)$, the expectation values of x, p are calculated as follow:

$$\langle x\rangle = \langle xS_\alpha, S_\alpha\rangle = \sqrt{\frac{\hbar}{2mw}}\left\langle (a^+ + a^-)S_\alpha, S_\alpha\right\rangle = \sqrt{\frac{\hbar}{2mw}}(\left\langle a^+ S_\alpha, S_\alpha\right\rangle + \left\langle a^- S_\alpha, S_\alpha\right\rangle) \tag{46}$$

$$= \sqrt{\frac{\hbar}{2mw}}(\left\langle S_\alpha, a^- S_\alpha\right\rangle + \langle \alpha S_\alpha, S_\alpha\rangle) = \sqrt{\frac{\hbar}{2mw}}(\langle S_\alpha, \alpha S_\alpha\rangle + \alpha) = \sqrt{\frac{\hbar}{2mw}}(\alpha^* + \alpha) \tag{47}$$

$$= \sqrt{\frac{\hbar}{2mw}} \cdot 2\text{Re}(\alpha) = \sqrt{\frac{\hbar}{2mw}} \cdot 2u \tag{48}$$

$$\langle p\rangle = \langle \hat{p}S_\alpha, S_\alpha\rangle = \sqrt{\frac{\hbar mw}{2}}\left\langle (a^+ - a^-)S_\alpha, S_\alpha\right\rangle = \sqrt{\frac{\hbar mw}{2}}(\left\langle a^+ S_\alpha, S_\alpha\right\rangle - \left\langle a^- S_\alpha, S_\alpha\right\rangle) \tag{49}$$

$$= \sqrt{\frac{\hbar mw}{2}}(\left\langle S_\alpha, a^- S_\alpha\right\rangle - \langle \alpha S_\alpha, S_\alpha\rangle) = \sqrt{\frac{\hbar mw}{2}}(\langle S_\alpha, \alpha S_\alpha\rangle - \alpha) = \sqrt{\frac{\hbar mw}{2}}(\alpha^* - \alpha) \tag{50}$$

$$= -\sqrt{\frac{\hbar mw}{2}} \cdot 2\text{Im}(\alpha) = -\sqrt{\frac{\hbar mw}{2}} \cdot 2v \tag{51}$$

(b) The expectation avlues of x^2, p^2 rae calculated as follow:

$$\left\langle x^2\right\rangle = \left\langle \hat{x}^2 S_\alpha, S_\alpha\right\rangle = \frac{\hbar}{2mw}\left\langle ((a^+)^2 + \text{id} + 2a^+a^- + (a^-)^2)S_\alpha, S_\alpha\right\rangle \tag{52}$$

$$= \frac{\hbar}{2mw}(\left\langle (a^+)^2 S_\alpha, S_\alpha\right\rangle + \langle \text{id}S_\alpha, S_\alpha\rangle + 2\left\langle a^+a^- S_\alpha, S_\alpha\right\rangle + \left\langle (a^-)^2 S_\alpha, S_\alpha\right\rangle) \tag{53}$$

$$= \frac{\hbar}{2mw}(\left\langle S_\alpha, (a^-)^2 S_\alpha\right\rangle + 1 + 2\left\langle a^- S_\alpha, a^- S_\alpha\right\rangle + \left\langle \alpha^2 S_\alpha, S_\alpha\right\rangle) \tag{54}$$

$$= \frac{\hbar}{2mw}(\left\langle S_\alpha, \alpha^2 S_\alpha\right\rangle + 1 + 2\langle \alpha S_\alpha, \alpha S_\alpha\rangle + \alpha^2) \tag{55}$$

$$= \frac{\hbar}{2mw}((\alpha^2)^* + 1 + 2\alpha\alpha^* + \alpha^2) = \frac{\hbar}{2mw}(1 + 2|\alpha|^2 + 2\text{Re}(\alpha^2)) \tag{56}$$

$$= \frac{\hbar}{2mw}(1 + 2u^2 + 2v^2 + 2(u^2 - v^2)) = \frac{\hbar}{2mw}(1 + 4u^2) \tag{57}$$

$$\langle p\rangle = \left\langle \hat{p}^2 S_\alpha, S_\alpha\right\rangle = -\frac{\hbar mw}{2}\left\langle ((a^+)^2 - \text{id} - 2a^+a^- + (a^-)^2)S_\alpha, S_\alpha\right\rangle \tag{58}$$

$$= -\frac{\hbar mw}{2}(\left\langle (a^+)^2 S_\alpha, S_\alpha\right\rangle - \langle \text{id}S_\alpha, S_\alpha\rangle - 2\left\langle a^+a^- S_\alpha, S_\alpha\right\rangle + \left\langle (a^-)^2 S_\alpha, S_\alpha\right\rangle) \tag{59}$$

$$= -\frac{\hbar mw}{2}(\left\langle S_\alpha, (a^-)^2 S_\alpha\right\rangle - 1 - 2\left\langle a^- S_\alpha, a^- S_\alpha\right\rangle + \left\langle \alpha^2 S_\alpha, S_\alpha\right\rangle) \tag{60}$$

$$= -\frac{\hbar mw}{2}(\left\langle S_\alpha, \alpha^2 S_\alpha\right\rangle - 1 - 2\langle \alpha S_\alpha, \alpha S_\alpha\rangle + \alpha^2) \tag{61}$$

$$= -\frac{\hbar mw}{2}((\alpha^2)^* - 1 - 2\alpha\alpha^* + \alpha^2) = -\frac{\hbar mw}{2}(2\text{Re}(\alpha^2) - 1 - 2|\alpha|^2) \tag{62}$$

$$= -\frac{\hbar mw}{2}(2u^2 - 2v^2 - 1 - 2(u^2 + v^2)) = \frac{\hbar mw}{2}(1 + 4v^2) \tag{63}$$

(c) Based on part (a) and (b), the product of uncertainty square is given as follow:

$$\sigma_p^2\sigma_x^2 = (\langle p^2\rangle - \langle p\rangle^2)(\langle x^2\rangle - \langle x\rangle^2) \tag{64}$$
$$= \left(\frac{\hbar m w}{2}(1+4v^2) - \left(-\sqrt{\frac{\hbar m w}{2}}\cdot 2v\right)^2\right)\left(\frac{\hbar}{2mw}(1+4u^2) - \left(\sqrt{\frac{\hbar}{2mw}}\cdot 2u\right)^2\right) \tag{65}$$
$$= \frac{\hbar m w}{2}(1+4v^2-4v^2)\cdot\frac{\hbar}{2mw}(1+4u^2-vu^2) = \frac{\hbar^2}{4} \tag{66}$$

Hence, it shows that $\sigma_p\sigma_x = \sqrt{\frac{\hbar^2}{4}} = \frac{\hbar}{2}$, which Uncertainty Principle is saturated.

(d) Express $S_\alpha(x) = \sum_{n=0}^{\infty} c_n\psi_n(x)$. Recall that $\alpha^-\psi_n(x) = \sqrt{n}\psi_{n-1}(x)$ and $a^- psi_0(x) = 0$, hence acting this linear operator on both sides, it derives:

$$\sum_{n=0}^{\infty}\alpha c_n\psi_n(x) = \alpha S_\alpha(x) = a^- S_\alpha(x) = c_0 a^-\psi_0(x) + \sum_{n=1}^{\infty} c_n a^-\psi_n(x) = \sum_{n=1}^{\infty} c_n\sqrt{n}\psi_{n-1}(x) \tag{67}$$
$$= \sum_{n=0}^{\infty} c_{n+1}\sqrt{n+1}\psi_n(x) \tag{68}$$

Hence, each coefficient must match, showing $\alpha c_n = c_{n+1}\sqrt{n+1}$ for all $n\in\mathbb{N}$, or $c_{n+1} = \frac{\alpha}{\sqrt{n+1}}c_n$.

Using induction we'll prove that $c_n = \frac{\alpha^n}{\sqrt{n!}}c_0$: For base case $n=1$, one has $c_1 = \frac{\alpha}{\sqrt{0+1}}c_0 = \frac{\alpha}{\sqrt{1!}}c_0$. Now, suppose for given $n\in\mathbb{N}$ one has $c_n = \frac{\alpha^n}{\sqrt{n!}}c_0$, then for the case $(n+1)\in\mathbb{N}$, one hase $c_{n+1} = \frac{\alpha}{\sqrt{n+1}}c_n = \frac{\alpha}{\sqrt{n+1}}\cdot\frac{\alpha^n}{\sqrt{n!}}c_0 = \frac{\alpha^{n+1}}{\sqrt{(n+1)!}}c_0$, hence the induction is done, showing that $c_n = \frac{\alpha^n}{\sqrt{n!}}c_0$ for all $n\in\mathbb{N}$.

(e) For normalization condition, we have the following:

$$1 = \langle S_\alpha, S_\alpha\rangle = \left\langle\sum_{n=0}^{\infty} c_n\psi_n, \sum_{m=0}^{\infty} c_m\psi_m\right\rangle = \sum_{(n,m)\in\mathbb{N}^2} c_n c_m^*\langle\psi_n,\psi_m\rangle \tag{69}$$

Rcall that from orthonormality, one hast $\langle\psi_n,\psi_m\rangle = \delta_{nm}$, which eliminates every pair of (n,m) where $n\neq m$. Hence, it becomes the follow:

$$1 = \sum_{n=0}^{\infty} c_n c_n^*\langle\psi_n,\psi_n\rangle = \sum_{n=0}^{\infty}|c_n|^2 \tag{70}$$

Using the formula in the previous part, c_0 satisfies:

$$1 = \sum_{n=0}^{\infty}|c_n|^2 = \sum_{n=0}^{\infty}|c_0|^2\frac{|\alpha|^{2n}}{n!} = |c_0|^2 e^{|\alpha|^2} \implies |c_0| = e^{-|\alpha|^2/2} \tag{71}$$

For easier interpretation, one can choose $c_0\in\mathbb{R}$, hence $c_0 = e^{-|\alpha|^2/2}$ satisfies the normalization condition.

(f) If consider $S_\alpha(x,t) = \sum_{n=0}^{\infty} c_n\psi_n(x)e^{-iE_n t/\hbar}$, recall that for all $n\in\mathbb{N}$, $E_n = \left(n+\frac{1}{2}\right)\hbar w$, hence $E_{n+1} = \left(n+1+\frac{1}{2}\right)\hbar w = E_n + \hbar w$. Hence, if acting α^- on $S_\alpha(x,t)$ (which is an operator involving only x, $\frac{\partial}{\partial x}$, hence has no effect on the time evolution), one gets:

$$a^- S_\alpha(x,t) = c_0(a^-\psi_0(x))e^{-iE_0t/\hbar} + \sum_{n=1}^{\infty}\frac{\alpha^n}{\sqrt{n!}}c_0\cdot(a^-\psi_n(x))e^{-iE_nt/\hbar} = \sum_{n=1}^{\infty}\frac{\alpha^n}{\sqrt{n!}}c_0\cdot\sqrt{n}\psi_{n-1}(x)e^{-iE_nt/\hbar} \tag{72}$$

$$= \sum_{n=1}^{\infty}\frac{\alpha^n}{\sqrt{(n-1)!}}c_0\psi_{n-1}(x)e^{-iE_nt/\hbar} = \sum_{n=0}^{\infty}\frac{\alpha^{n+1}}{\sqrt{n!}}c_0\psi_n(x)e^{-iE_{n+1}t/\hbar} \tag{73}$$

$$= \alpha\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}c_0\psi_n(x)e^{-i(E_n+\hbar w)t/\hbar} = \alpha e^{-iwt}\sum_{n=0}^{\infty}c_n\psi_n(x)e^{-iE_nt/\hbar} = \alpha e^{-iwt}\cdot S_\alpha(x,t) \tag{74}$$

Hence, with $\alpha(t) := \alpha e^{-iwt}$, one has $a^- S_\alpha(x,t) = \alpha(t)S_\alpha(x,t)$.

So, regardless of the time $t \in \mathbb{R}$, since $\alpha(t) = \alpha e^{-iwt}$ is a nonzero complex number independent of x, it implies $S_\alpha(x,t)$ remains as an eigenvector of a^-, and at each time $t \in \mathbb{R}$ with corresponding eigenvalue αe^{-iwt}, so it stays coherent.

Also, notice that the proof in part (a),(b),(c) only requires α to be a complex number (since regardless of $\alpha = u+iv$, the inequality in part (c) always holds, as long as $S_\alpha(x,t) \neq 0$), hence as long as $S_\alpha(x,t)$ itself is a normalizable state that's an eigenvector of α^-, regardless of its eigenvalue α corresponding to α^-, it still saturates Uncertainty Principle.

(g) Finally, based on the above claim, since $\psi_0 \neq 0$, while $a^-\psi_0 = 0\cdot\psi_0$, then set $\alpha = 0$, $S_\alpha(x,t) = \psi_0(x,t)$ is a coherent state.

□